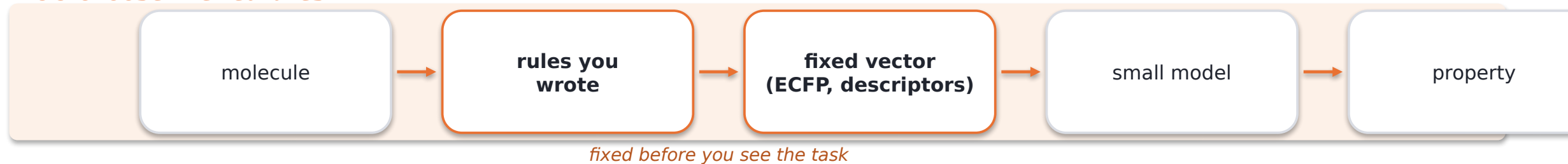


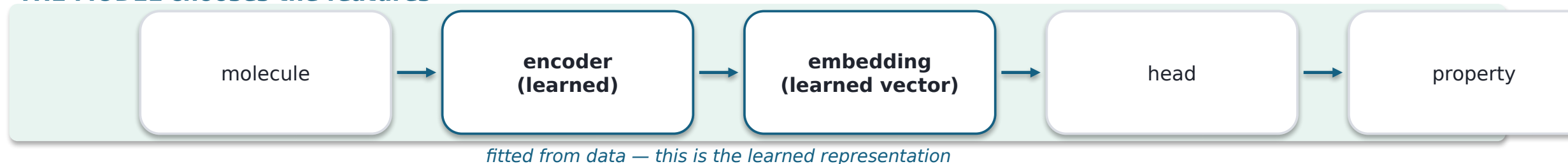
Learned Representations

Letting the model choose the features

YOU choose the features



THE MODEL chooses the features



Plan

- **Why learn a representation?** the label bottleneck (10 min)
- **Unsupervised learning:** clustering, chemical space maps, what they do and do not tell you (20 min)
- **Autoencoders and VAEs:** compression, latent spaces, and an honest look at molecular generation (20 min)
- — break —
- **Self-supervised pre-training:** masked prediction and contrastive learning (15 min)
- **Transfer learning:** frozen, partial, full — and where it genuinely pays (15 min)

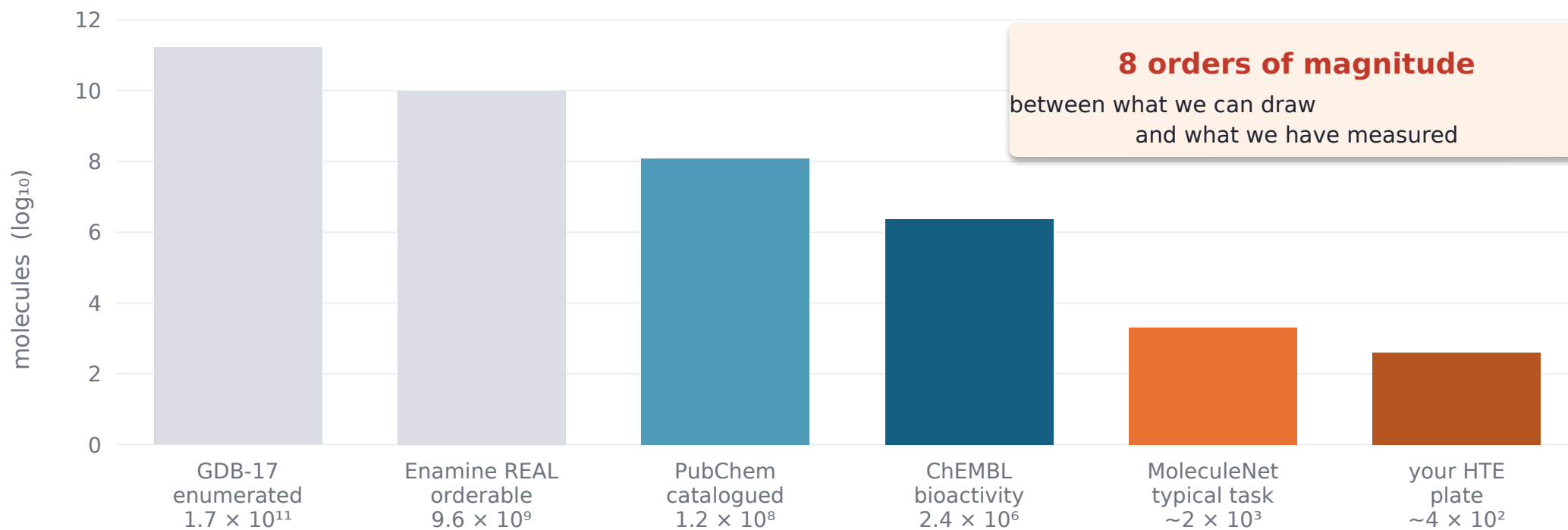
Tutorial at 15:15 — you will build the map and measure the crossover yourself, on real data.

01

Why learn a representation?

- Hand-crafted vs learned features
- The label bottleneck
- Three kinds of supervision

The bottleneck is labels, not molecules



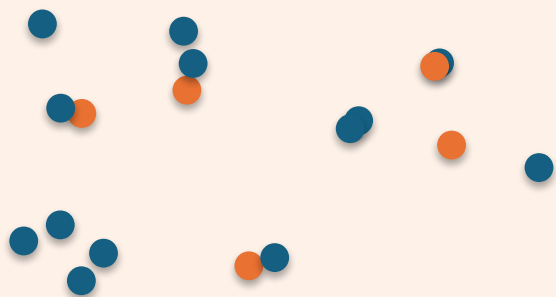
Molecules are free. **Labels are the scarce resource** — everything today spends the first to make up for the second.

GDB-17: Ruddigkeit et al., J. Chem. Inf. Model. 2012, 52, 2864 · Enamine REAL (2026) · PubChem · ChEMBL

Three ways to learn — only one of them needs you

SUPERVISED

learn $x \rightarrow y$

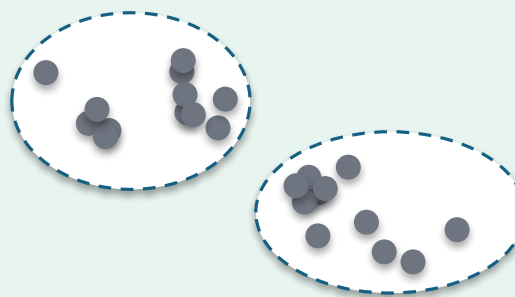


every point carries a label

labels are expensive

UNSUPERVISED

learn the shape of x

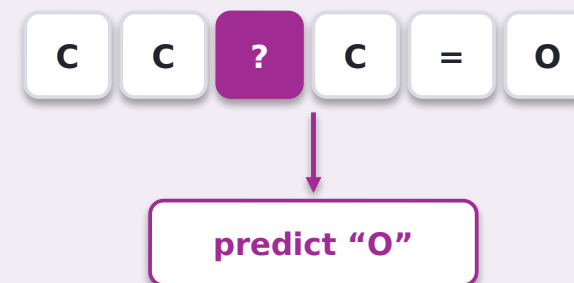


no labels at all — only structure

unlimited data

SELF-SUPERVISED

learn $x \rightarrow \text{part of } x$



the data labels itself

supervised signal, no annotation

Self-supervised learning turns unlabelled molecules into a training signal. That is the bridge from 10^9 molecules to a trainable model.

02

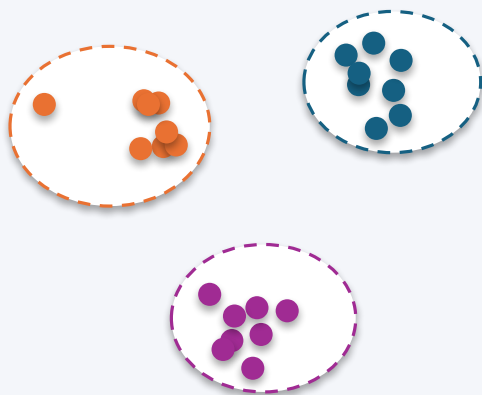
Unsupervised learning

- Clustering, dimensionality reduction, density
- Chemical space maps — and how to read them honestly
- Case study: the periodic table from text alone

Unsupervised learning does three jobs

1 · CLUSTERING

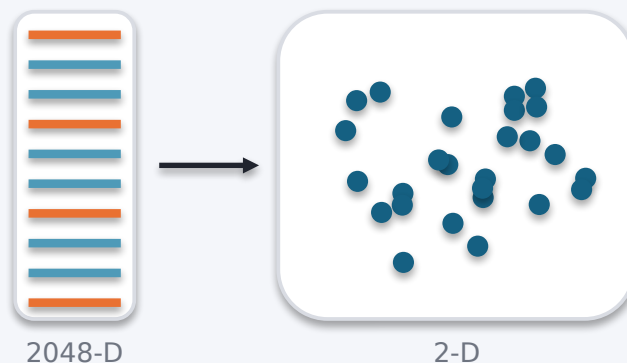
which molecules belong together?



groups, no coordinates

2 · DIMENSIONALITY REDUCTION

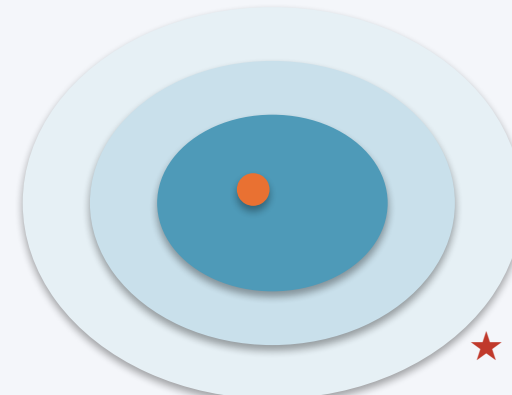
can I see 2048 dimensions?



coordinates you can plot

3 · DENSITY & GENERATION

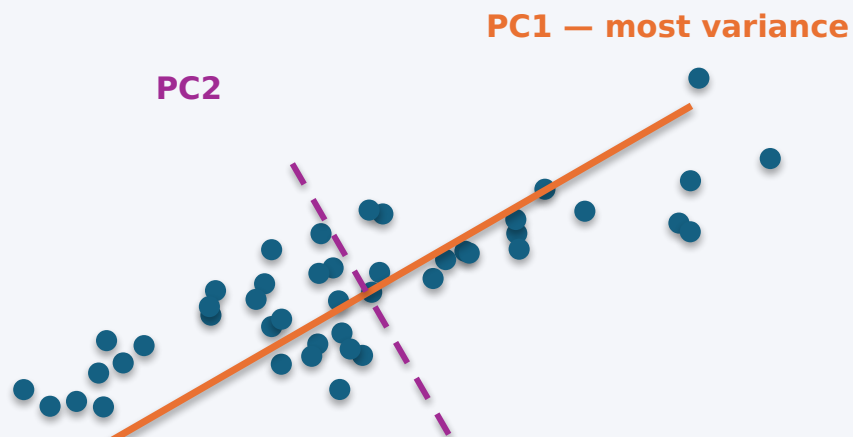
what does a plausible molecule look like?



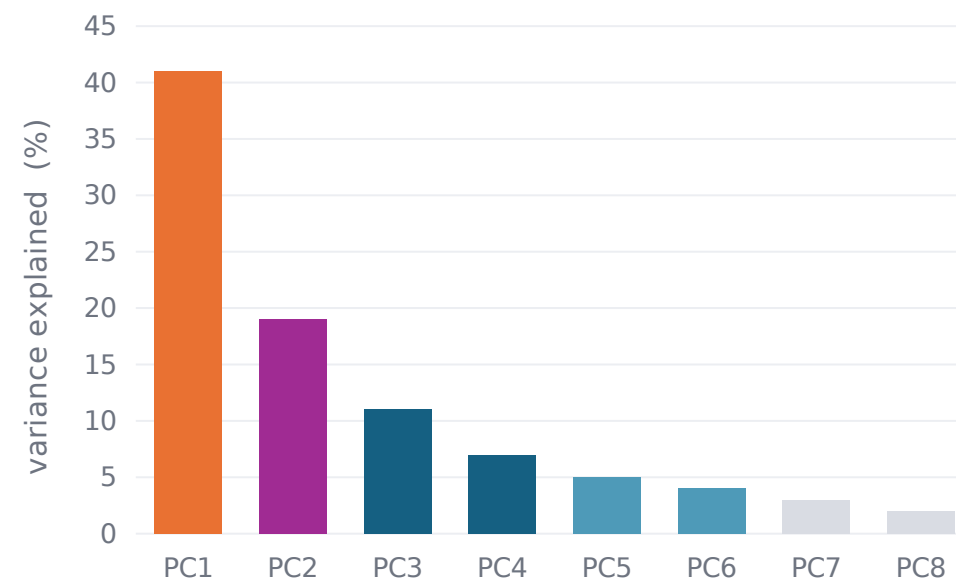
sample new points from the model

Decide which job you want **before** you open the notebook — a UMAP is not a clustering, and a clustering is not a generative model.

PCA: a rotation, and an honest one



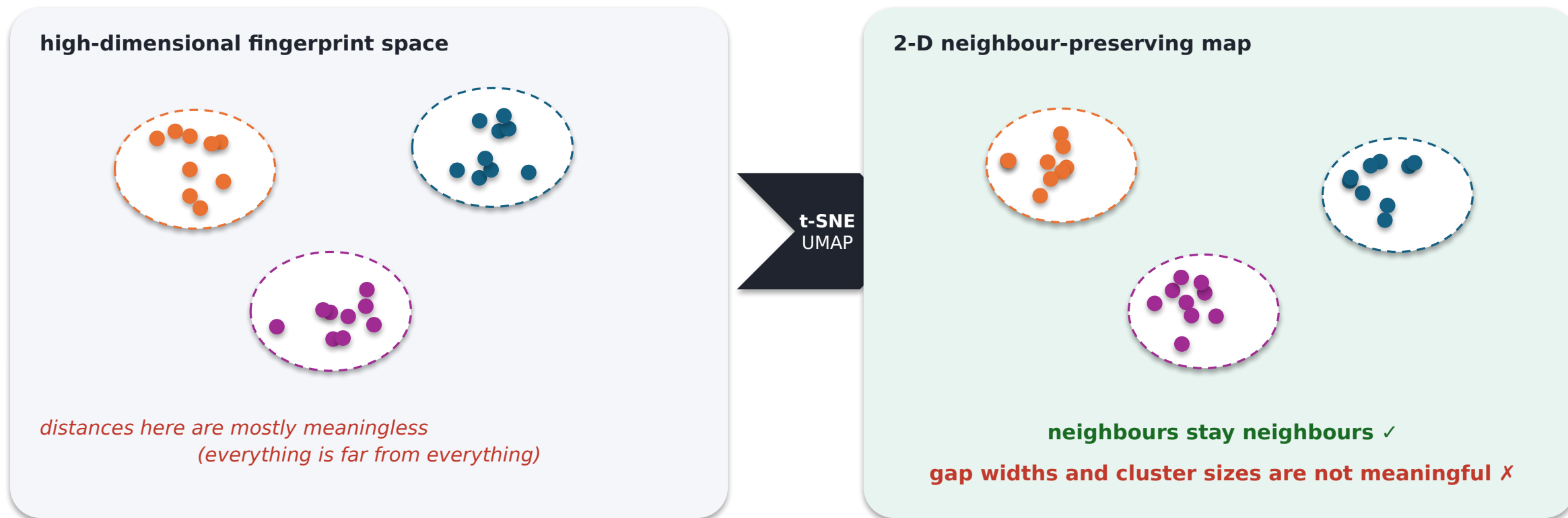
a rotation, nothing more



2 components $\approx$ 60% of the variance — the rest is not noise, it is chemistry

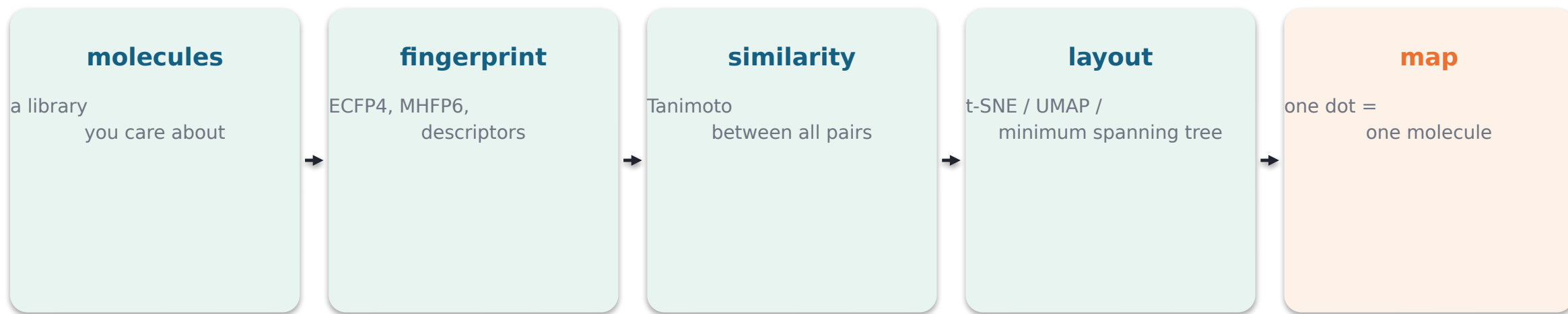
PCA is transparent and invertible, but **two components never show you fingerprint data** — the variance you cannot see is chemistry, not noise.

t-SNE and UMAP: keep the neighbours, lose the distances



The trade is deliberate: **neighbourhoods are preserved, distances are not**. That is the right trade for fingerprints — and a trap if you forget it.

A chemical space map is a chain of choices

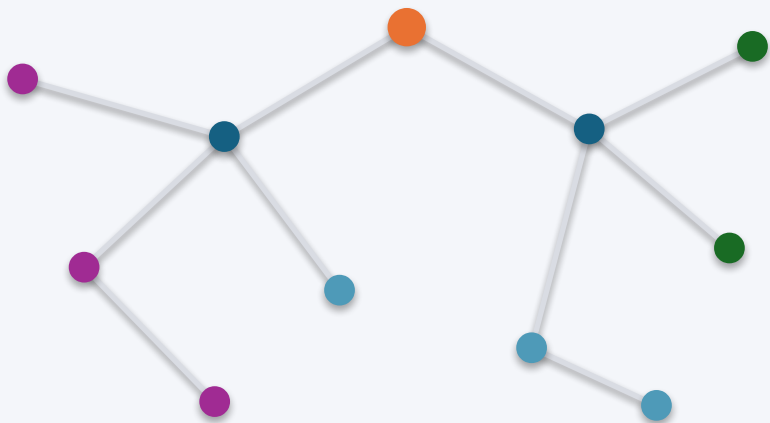


every choice in this chain changes the picture — the map is a property of your fingerprint, not of chemistry

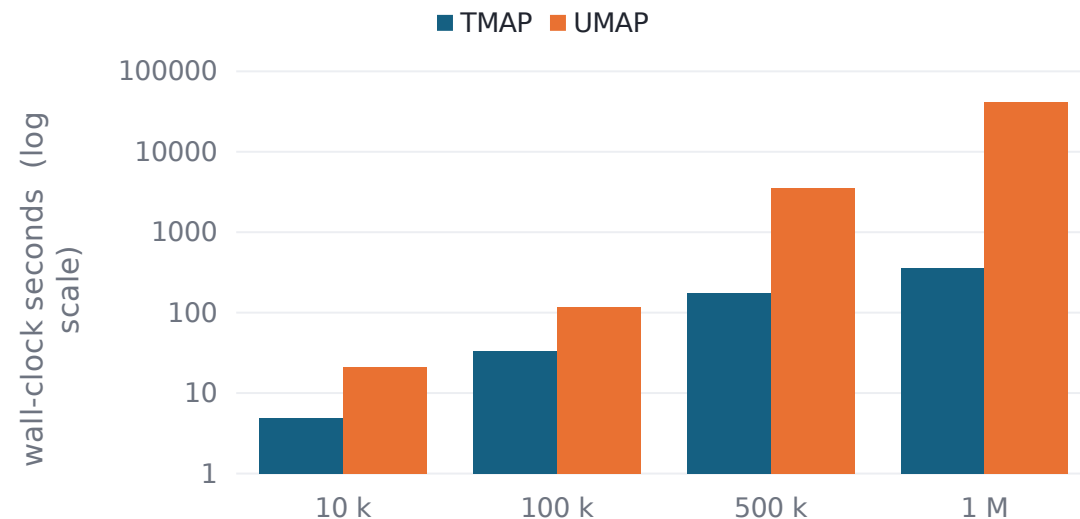
A chemical space map is **a picture of your fingerprint**, not a picture of chemistry. Change the fingerprint and the story changes.

At a million molecules, the algorithm matters

TMAP: lay out a minimum spanning tree, not a scatter



branches = series of analogues · no overplotting at 10^6 molecules

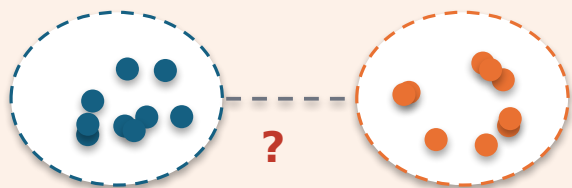


1 M molecules: TMAP 6 min vs UMAP 11.5 h (~117×)

TMAP: ~117× faster than UMAP at 10^6 molecules, and reproducible. Branches are analogue series — a chemically meaningful unit.

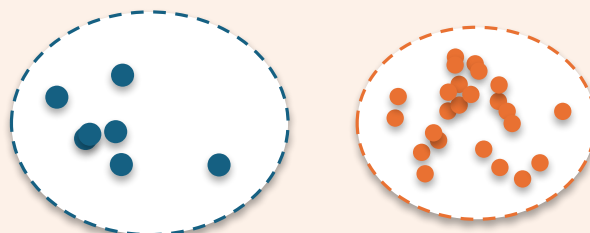
Three things your map is not telling you

X gap width means nothing



perplexity changes it

X cluster size means nothing

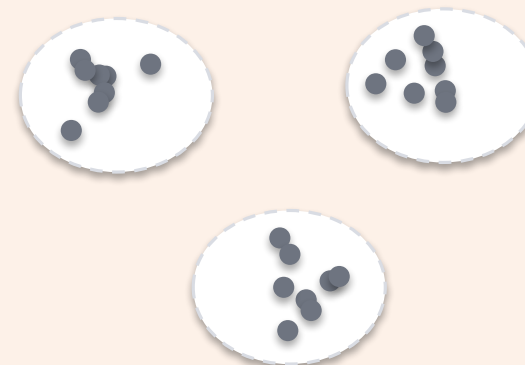


7 molecules

240 molecules

drawn almost the same size

X clusters appear in pure noise



input was uniform random

Never read clusters off a map. Cluster explicitly in the original space, then use the map only to display the result.

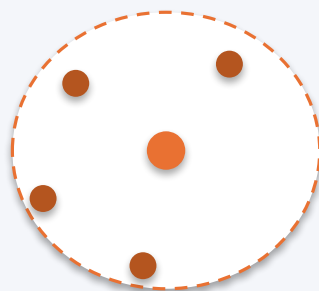
Butina clustering: the cheminformatics default

1 · count neighbours within t



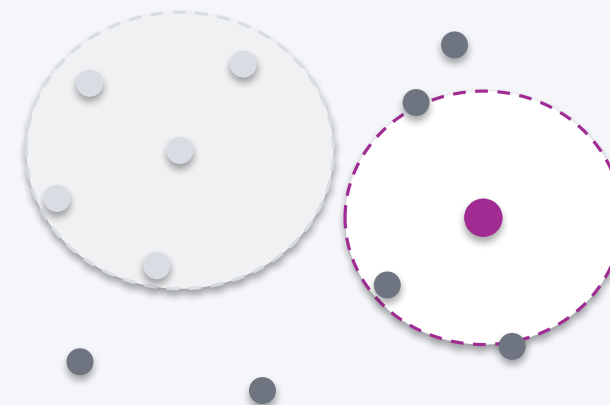
$t = \text{Tanimoto cut-off}$

2 · densest molecule claims its sphere



cluster centre + members

3 · repeat on what is left

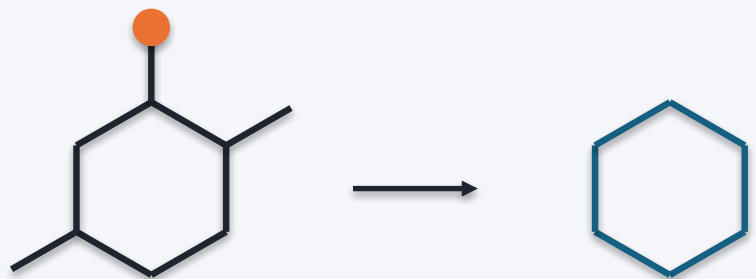


greedy, deterministic, no k to choose

No k to choose, deterministic, and the centres are real molecules. Run it at two or three cut-offs and check your conclusion survives.

Scaffolds: chemistry is far more clustered than it looks

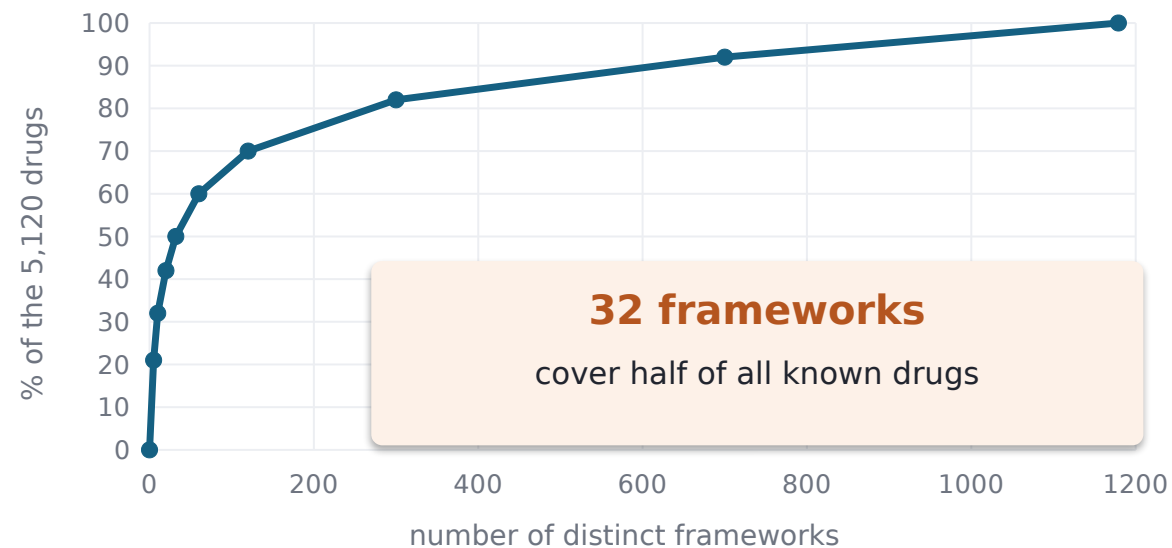
strip side chains → keep the framework



molecule

Bemis-Murcko
framework

**scaffold split = hold out whole frameworks,
not random molecules**

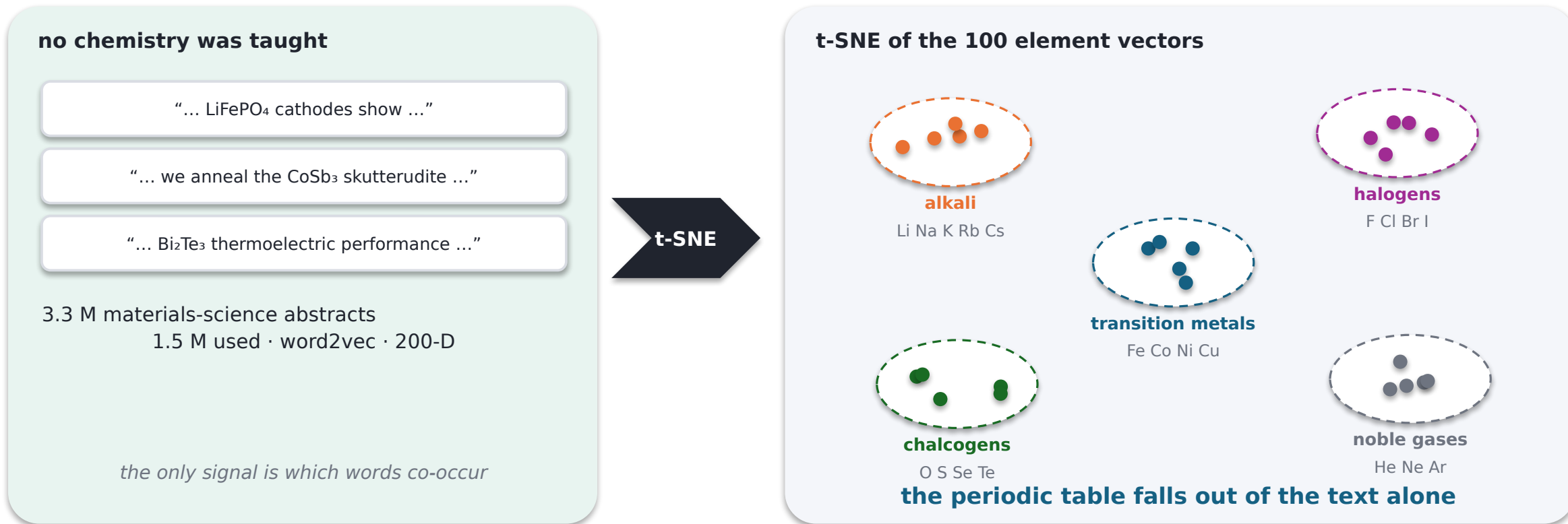


1,179 distinct frameworks among 5,120 drugs

32 frameworks cover half of all known drugs. This is why you must split by scaffold — and why your effective sample size is smaller than you think.

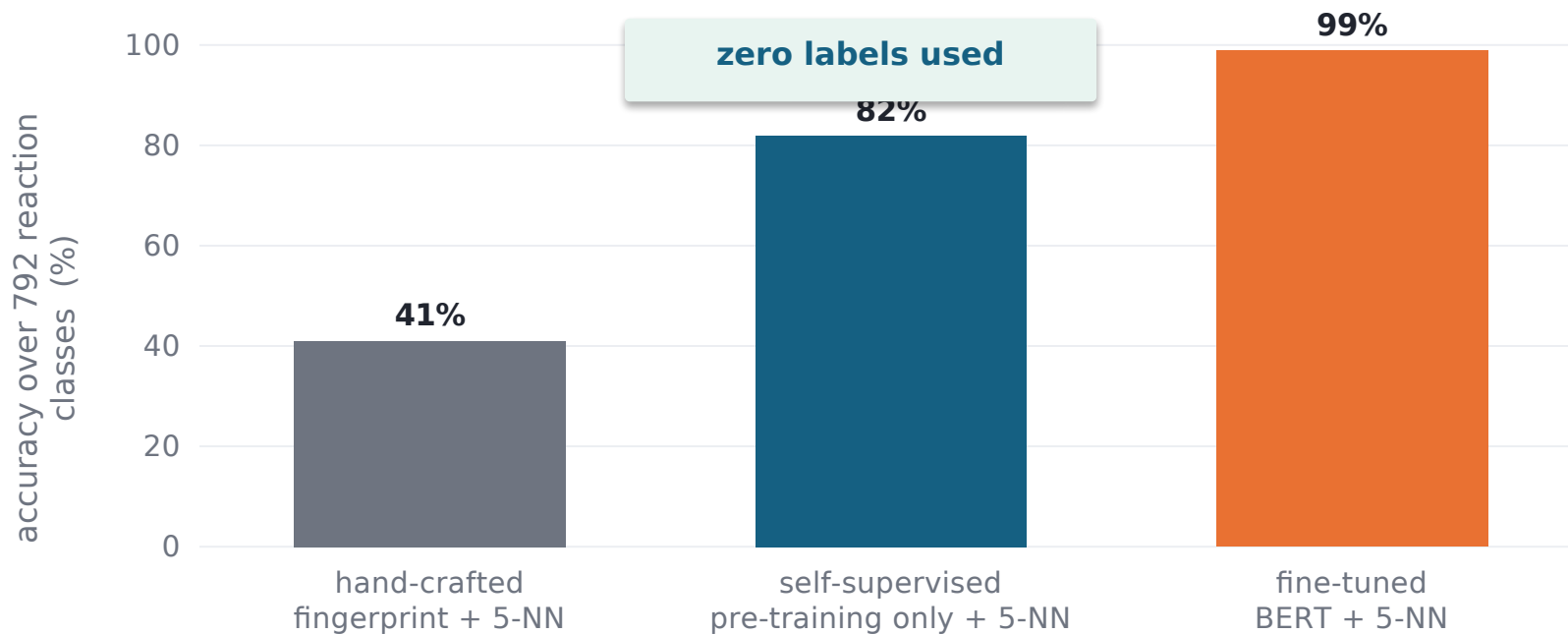
Bemis & Murcko, J. Med. Chem. 1996, 39, 2887 — 1,179 frameworks among 5,120 drugs

Case study: the periodic table, from text alone



Nobody taught it any chemistry. **Unsupervised learning on 3.3 M abstracts recovers the periodic table** — and predicts new thermoelectrics.

Case study: reaction classes, unsupervised



2.6 M unlabelled reactions in · reaction classes come out

Masked-language pre-training alone recovers most of the reaction taxonomy — 81.9% vs 41.0% for a hand-crafted fingerprint, with zero labels.

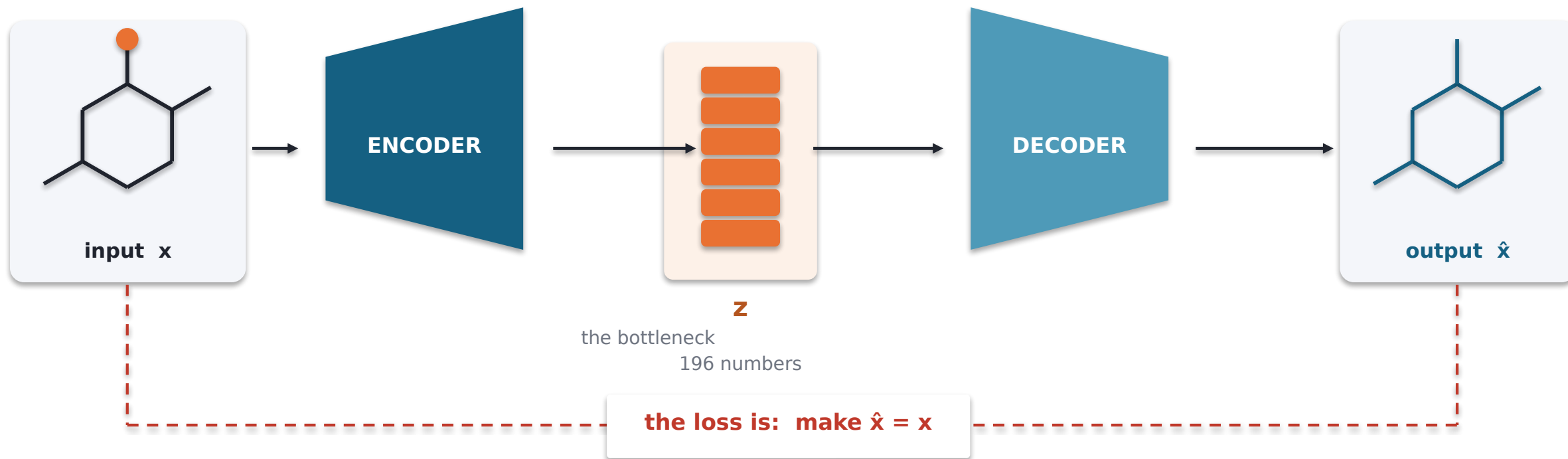
Schwaller et al., Nat. Mach. Intell. 2021, 3, 144 · counterpoint: Probst et al., Digital Discovery 2022, 1, 91 (DRFP)

03

Autoencoders and VAEs

- Compression as representation learning
- Why a plain autoencoder is not enough
- What VAEs are — and are not — good for

The autoencoder: learn by rebuilding the input

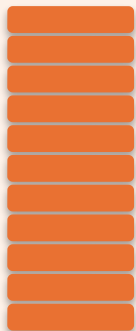


The molecule is its own label. **If 196 numbers can rebuild the molecule, those 196 numbers are a representation the model invented.**

The bottleneck is the whole trick

bottleneck too wide

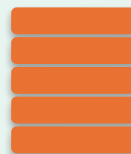
z is as big as x



the network copies the input
and learns nothing

bottleneck just right

$z \ll x$



it must keep ring systems,
functional groups, size —
and throw away the rest

bottleneck too narrow

$z \approx 2$

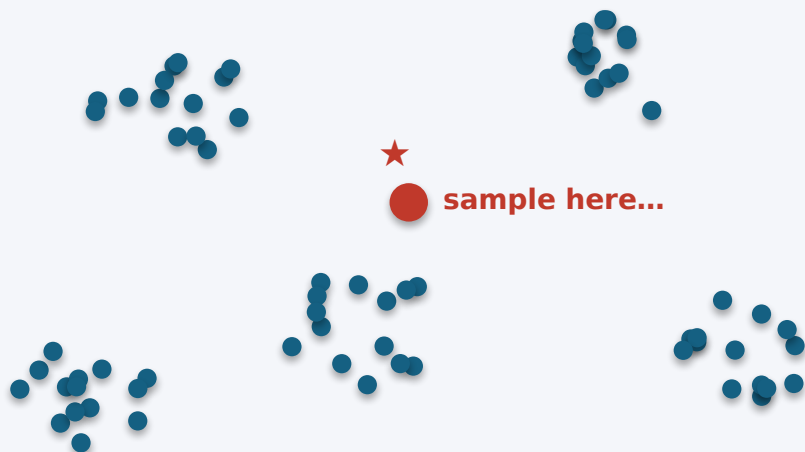


everything reconstructs
to benzene

compression is the whole trick: what survives the bottleneck is what the data says matters

The problem: a plain autoencoder learns islands

plain autoencoder — latent space



the model was never asked what lives between the islands

decode
→

what comes out

C1=CC=CC=C1C(=O)N(C)(C)(C)C ✗

c1ccc2cc(✗

CC(C)(C)(C)(C)C ✗

N#N#N=O ✗

C1CC1CC1CC1CC... ✗

invalid, unparseable, or chemically absurd

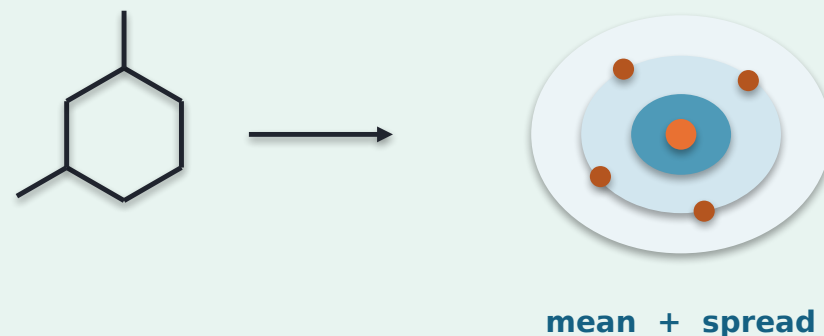
Nothing in the loss describes the space **between** the molecules — so the model never learns it, and sampling there produces garbage.

The VAE in one idea: encode a cloud, not a point

autoencoder: $x \rightarrow \text{one point}$



VAE: $x \rightarrow \text{a small cloud}$



training on samples from the cloud forces nearby points to decode to similar molecules — the gaps get filled

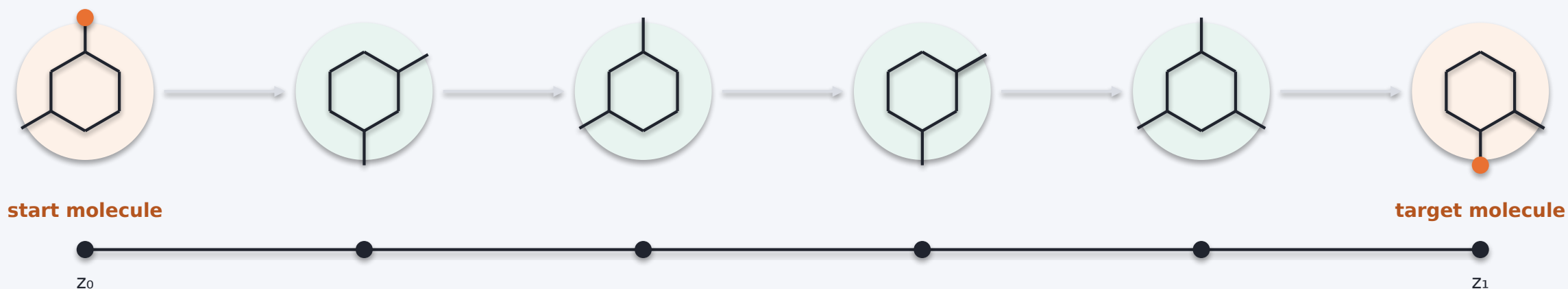
Two forces, and a dial between them



β is the whole design decision. Too small and you are back to islands; too large and the decoder ignores z entirely.

The payoff: a space you can walk through

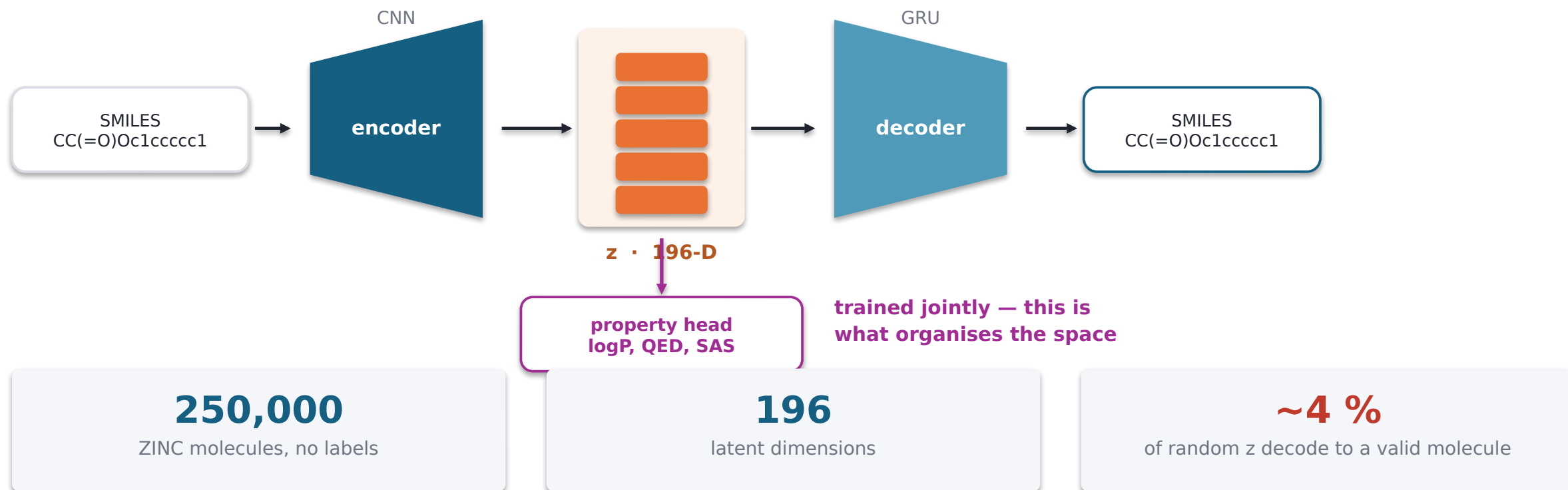
a straight line in z is a smooth series of molecules in chemistry



this is what people mean by “a continuous representation of molecules”

A VAE turns a **discrete** object into a **continuous** one — which is what every optimiser, gradient and Bayesian loop wants.

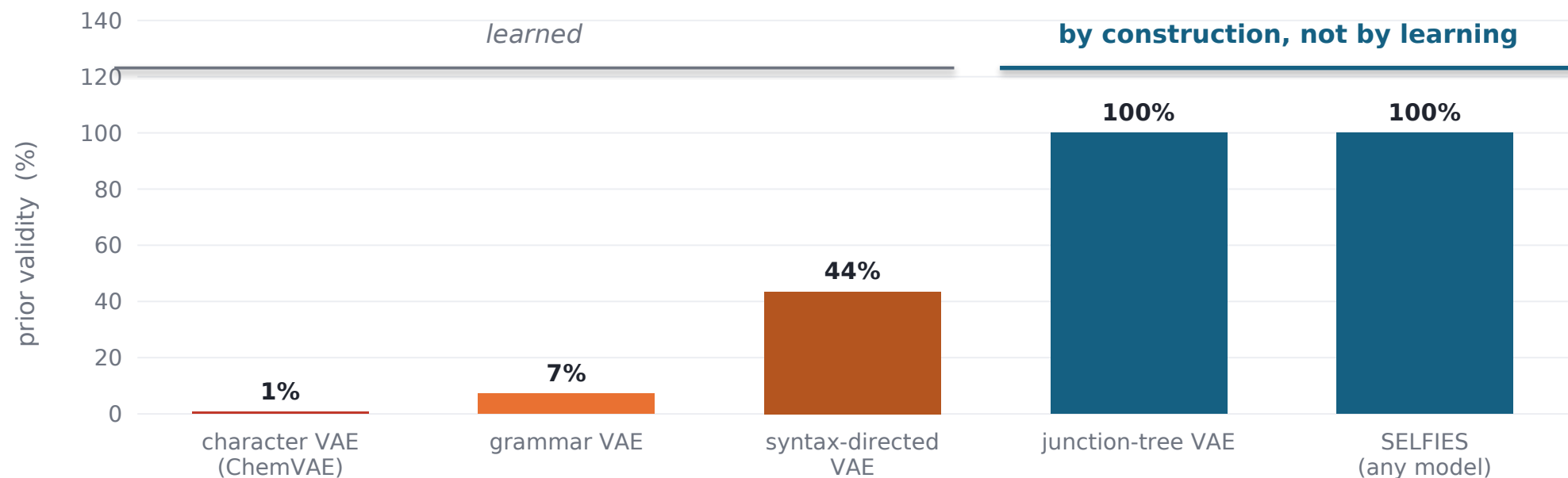
ChemVAE, 2018 — the paper that started it



Joint property training is the idea that survived: **give the encoder a whiff of the task and the latent space organises itself around it.**

Gómez-Bombarelli et al., ACS Cent. Sci. 2018, 4, 268 — validity figures are version- and definition-dependent; see notes

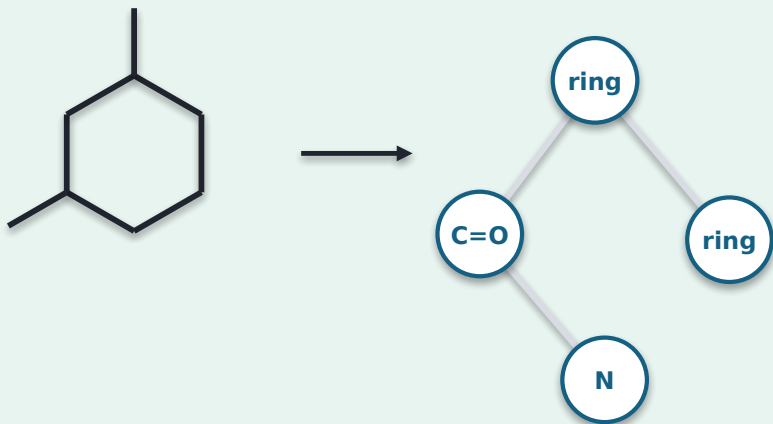
The validity arms race



The last two bars are won by construction, not by learning. The model did not get better at chemistry; the output format got stricter.

Two ways to make invalidity impossible

JT-VAE — assemble from a vocabulary of valid pieces

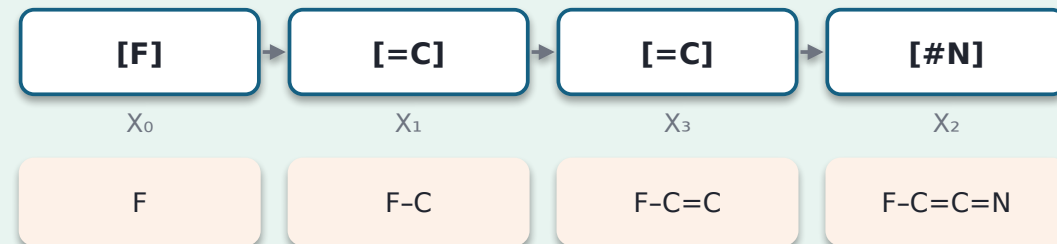


junction tree of substructures

780-piece vocabulary · illegal joins are masked out during decoding

100 % valid

SELFIES — a grammar that cannot fail



asked for a triple bond, valence allowed only a double → clamped, never rejected

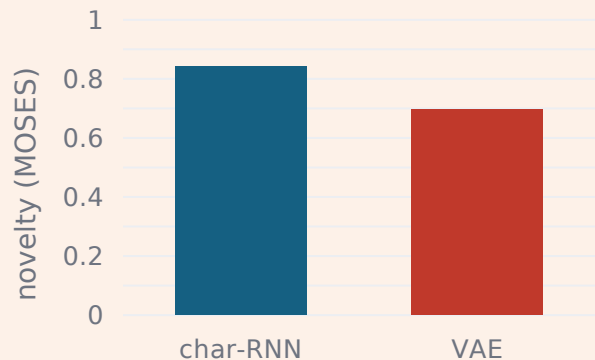
100 % valid · but only local valence is enforced —
not aromaticity, not ring strain, not stability

1 random mutation: SMILES 9.9 % → SELFIES 100 % valid

Both guarantee **syntactic** validity only. A 100%-valid molecule can still be one no chemist would draw.

The turn: validity was the easy metric

it memorises



31 % of what the VAE “designs”
is already in the training set

MOSES benchmark

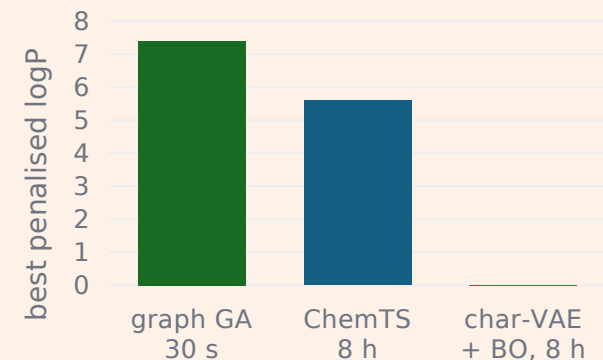
100 % valid $\neq$ good

model	valid %	FCD ↓
Molecule Chef	99.0	0.73
CGVAE	100.0	11.73
char-VAE	12.0	37.65

the model with perfect validity
matches the real distribution worst

FCD = distance to real chemistry

a genetic algorithm wins



30 seconds on a laptop beats
8 hours of VAE + Bayesian opt.

Jensen, Chem. Sci. 2019

Validity was easy, so it got measured; quality is hard, so it did not. A 30-second genetic algorithm beats 8 hours of VAE + Bayesian optimisation.

MOSES: Polykovskiy et al., Front. Pharmacol. 2020 · Molecule Chef: Bradshaw et al., NeurIPS 2019 · Jensen, Chem. Sci. 2019, 10, 3567

Keep the encoder, throw away the decoder

as a molecule generator

- ✗ beaten by a graph genetic algorithm
- ✗ beaten by an ordinary SMILES LSTM
- ✗ beaten by random screening of ZINC
- ✗ only ~30 % of proposals are synthesisable

*the field moved to diffusion, GAs and LLMs
(→ Thursday's lecture)*

as a representation learner

- ✓ a fixed-length vector for any molecule
- ✓ trained without a single label
- ✓ a smooth space you can optimise inside
- ✓ the encoder is reusable — that is the part that survived

keep the encoder, throw away the decoder

The decoder disappointed. The encoder became the foundation of everything after it — pretext task in, reusable encoder out.



Break

10 minutes — back for pre-training and transfer learning

04

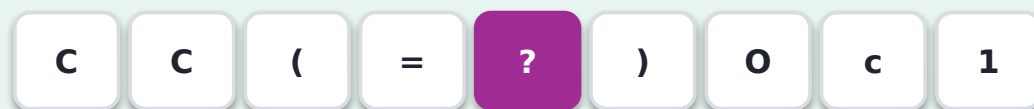
Self-supervised pre-training

- Masked prediction and contrastive learning
- Chemical language models at scale
- Pre-training graph neural networks

Two ways to make the data label itself

A · MASKED PREDICTION

hide part of the molecule, predict it back



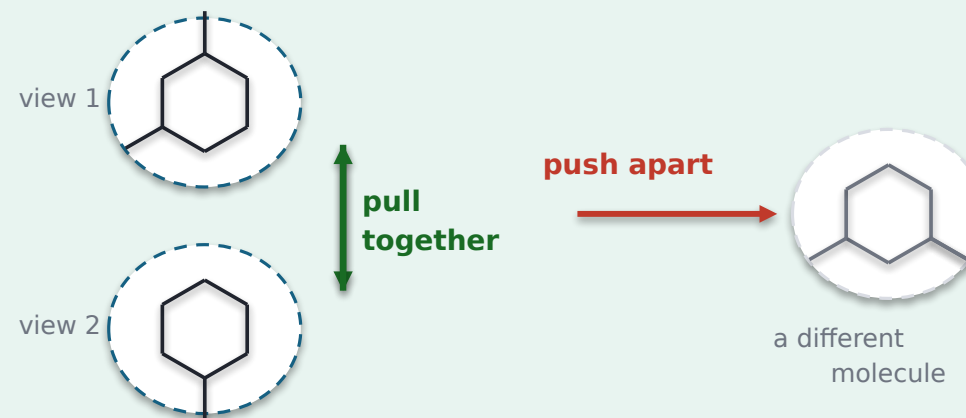
it was "O"

to fill the gap the model must learn valence,
ring closure and functional-group grammar

ChemBERTa · MolFormer · attribute masking on graphs

B · CONTRASTIVE

two damaged views of one molecule must agree



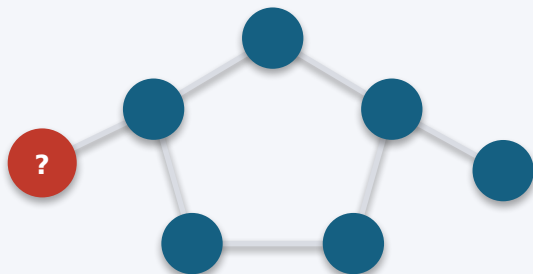
MolCLR · GraphCL · InfoMax

Masked prediction teaches **local chemistry**; contrastive teaches **what to ignore**. Both need only unlabelled molecules.

Choosing augmentations is choosing your chemistry

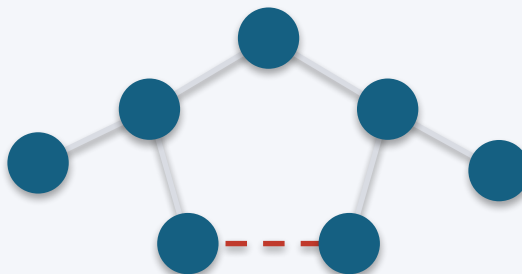
atom masking

*replace an atom type
with a dummy*



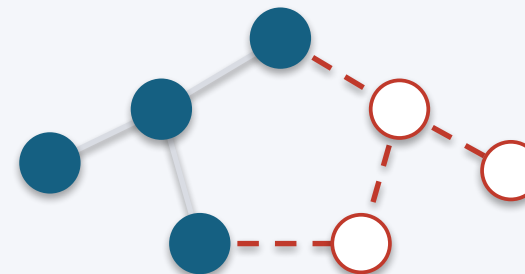
bond deletion

*drop a bond,
keep the atoms*



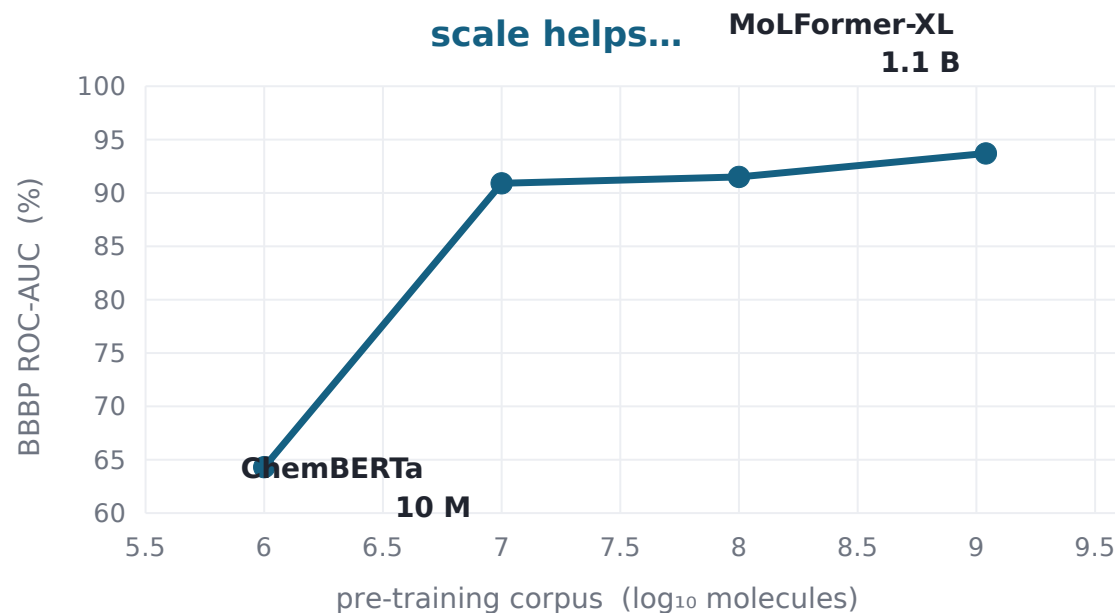
subgraph removal

*delete a whole
connected region*



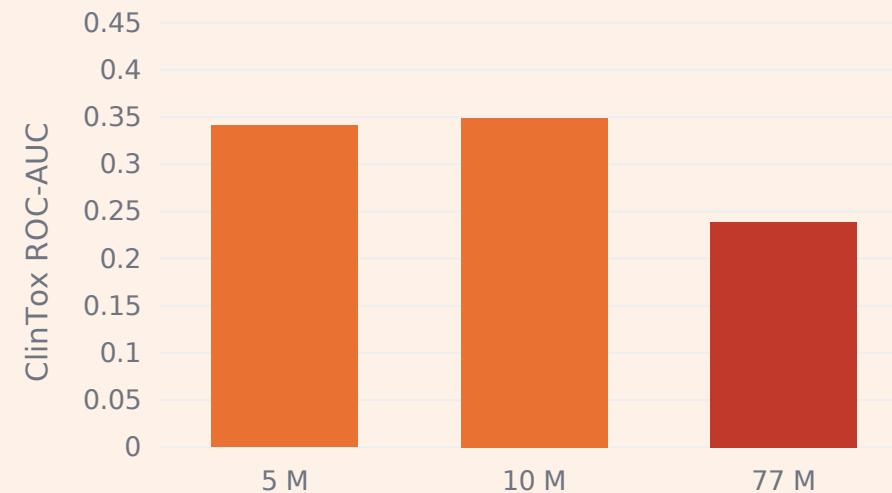
ablation: subgraph removal alone works best — combining all three destroys too much topology, and on BBBP it makes things worse

Scale helps — until it doesn't



1.1 B SMILES on 16 GPUs — linear attention made it affordable

...until it doesn't

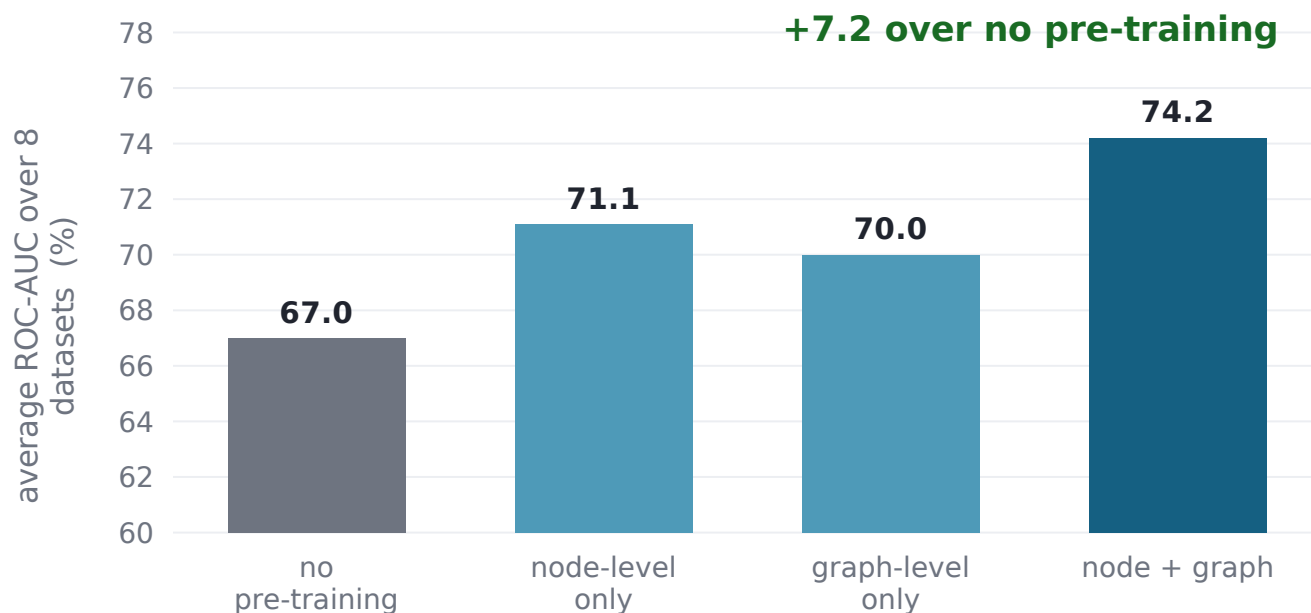


ChemBERTa-2, masked-LM: a 15× bigger corpus made this task worse

corpus composition matters more than corpus size

Corpus composition beats corpus size. A 15× bigger corpus made ClinTox substantially worse; a smaller, more diverse one beat a bigger, narrower one.

Pre-training GNNs: local and global, or negative transfer



the same recipe, different backbone



pre-training a weaker architecture actively hurts

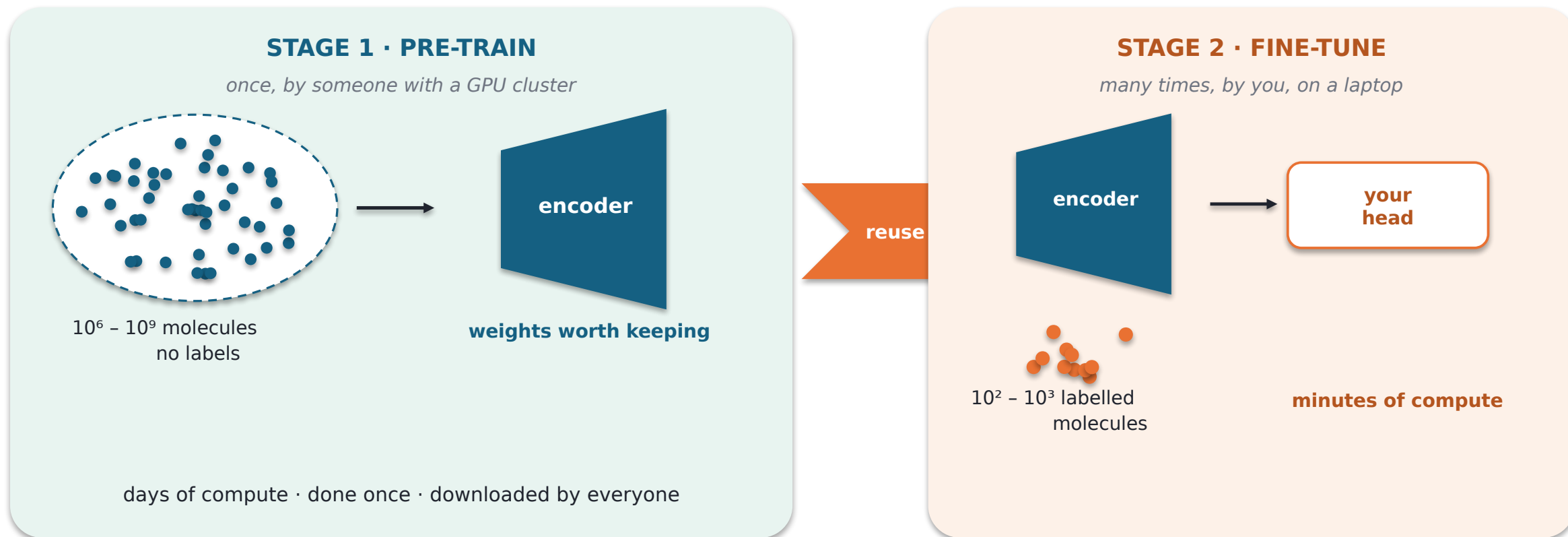
“We pre-trained it” guarantees nothing. **Pre-training a weaker architecture cost 6.5 points of ROC-AUC.**

05

Transfer learning

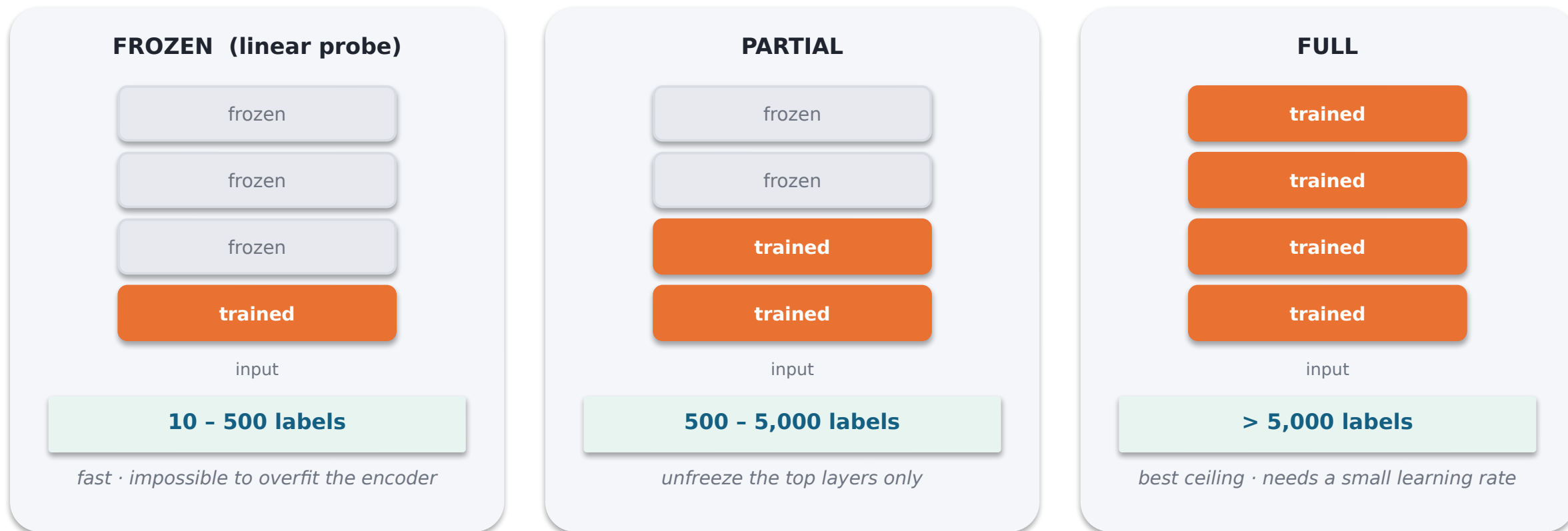
- Pre-train once, fine-tune many times
- Frozen, partial, full
- Where it genuinely pays

The recipe: pre-train once, fine-tune many times



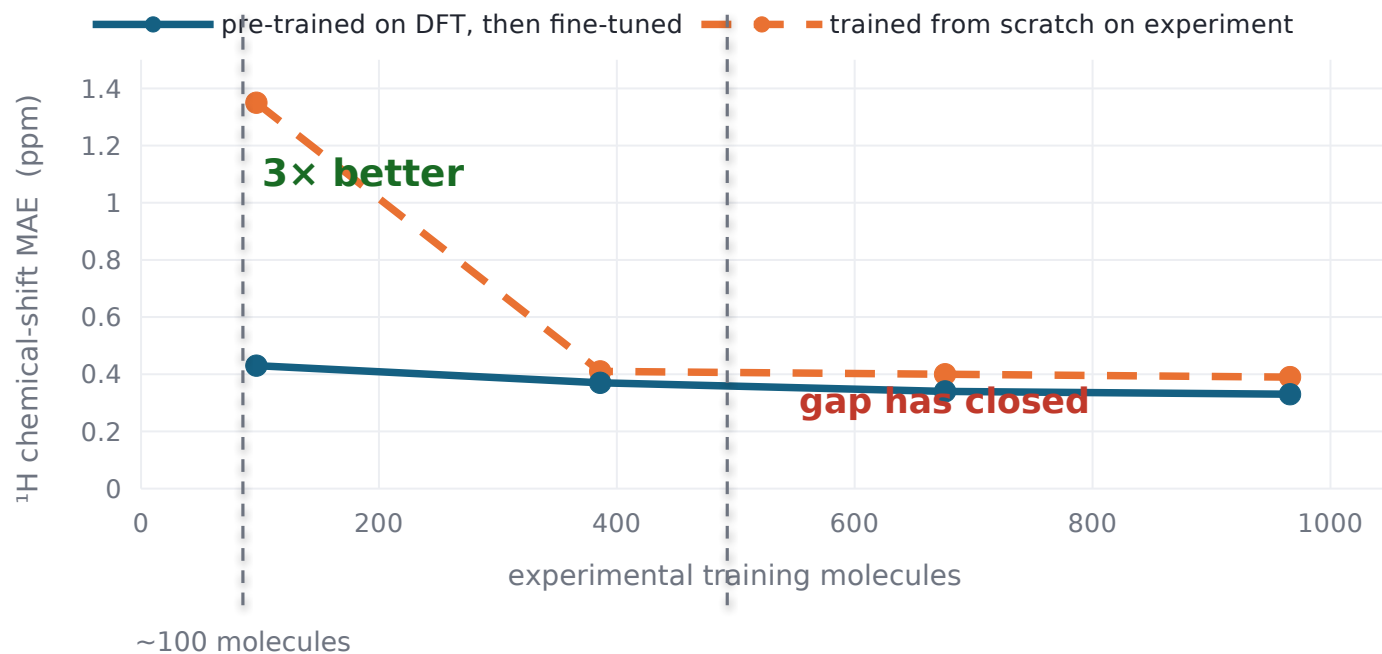
The expensive half is **amortised across the whole community**. You are buying a billion molecules' worth of chemistry for a coffee break.

Three knobs — pick by your dataset size



Always run the frozen probe first. Two minutes, no risk, and it tells you whether the representation contains anything useful for your task.

Where pre-training genuinely pays



the whole lecture in one plot

- Pre-training buys you data you do not have.
- Its value is largest when you have almost nothing.
- It decays to zero — and for ^{13}C it went negative — once you have ~1,000 labels.

Pre-training is a prior: worth 3× when you have ~100 labels, worth nothing by ~1,000, and occasionally worth less than nothing.

Han & Choi, *J. Phys. Chem. Lett.* 2021, 12, 3662 — ^1H MAE vs experimental training-set size (SI Table S2)

The same shape, in three different fields

REACTION YIELD

98

reactions are enough for a pre-trained reaction encoder to match DFT descriptors + RF

Buchwald-Hartwig HTE · Yield-BERT

INTERATOMIC POTENTIALS

3 %

of the data needed to specialise a foundation MLIP to liquid water, vs training from scratch

MatterSim · 17 M structures pre-trained

SUBLIMATION ENTHALPIES

tens

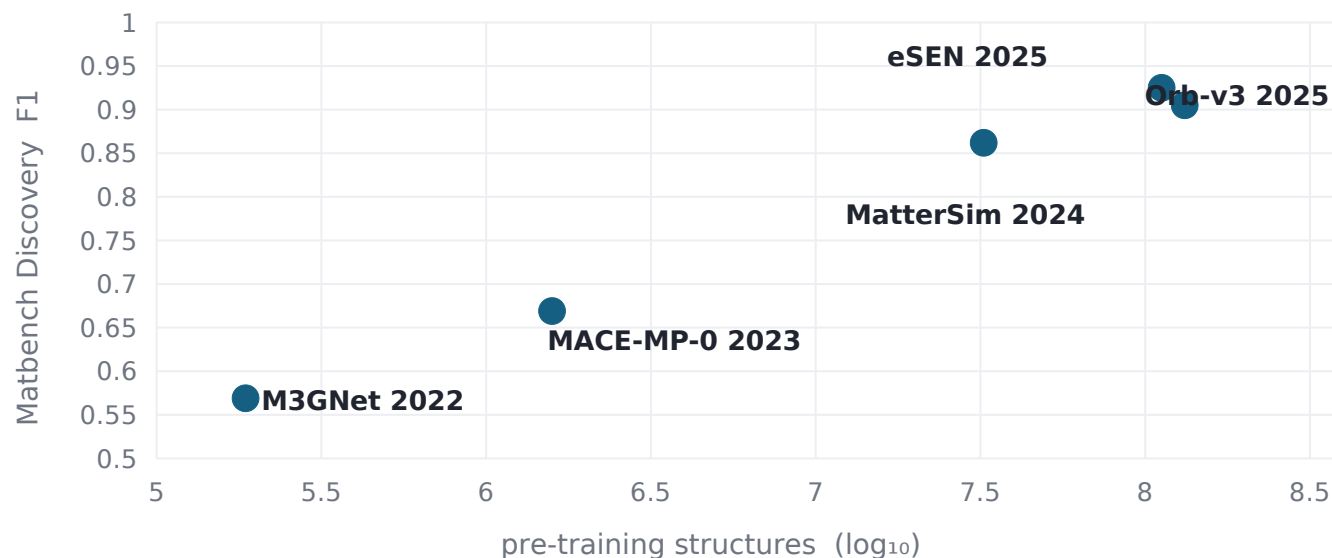
of training structures give sub-kJ/mol accuracy after fine-tuning MACE-MP-0

Kaur et al., Faraday Discuss. 2025

Different fields, different architectures, **the same shape** — the value of pre-training is concentrated where labels are most expensive.

Foundation models for atoms — the same play, at scale

~600x more training structures in three years



why chemists care

You will almost never train one of these.

You download it, then fine-tune on the 50–500 structures you can afford to compute for your own system.

Exactly the recipe from the last slide — applied to potentials instead of properties.

You will never train one of these — you will fine-tune one. Same recipe, applied to potentials instead of properties.

Matbench Discovery (matbench-discovery.materialsproject.org), values as of 11 Aug 2026 · Riebesell et al., Nat. Mach. Intell. 2025, 7, 836

06

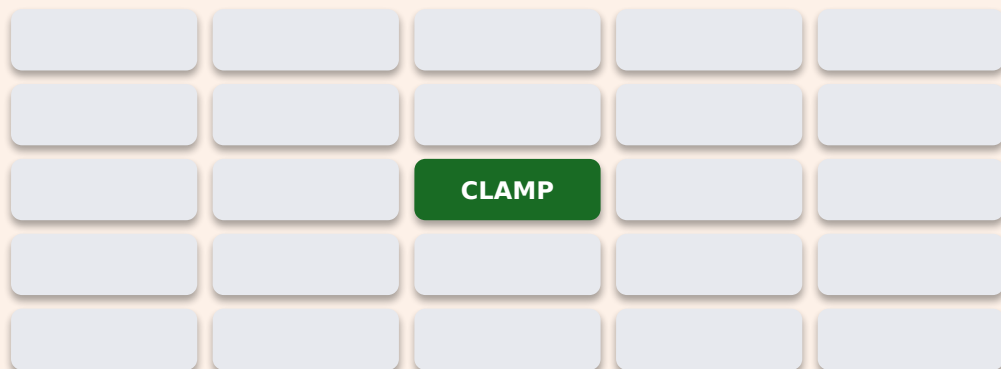
When it fails

- Negative transfer and the replication problem
- Activity cliffs
- The baseline you must always run

The uncomfortable benchmark

25 pre-trained models × 25 datasets

*re-evaluated with hierarchical Bayesian testing
instead of leaderboard means*



**one model beats the ECFP baseline significantly —
and it is itself fingerprint-based**

“

*nearly all neural models show
negligible or no improvement over
the baseline ECFP molecular
fingerprint.*

Praski, Adamczyk & Czech, arXiv:2508.06199 (2025)

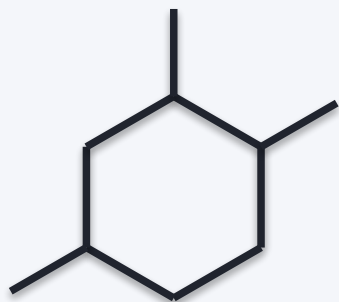
**Not a reason to give up on learned representations.
A reason to always run the ECFP + random-forest
baseline first.**

Always run ECFP + random forest first. If your model cannot beat it, you have learned something valuable for the price of two minutes.

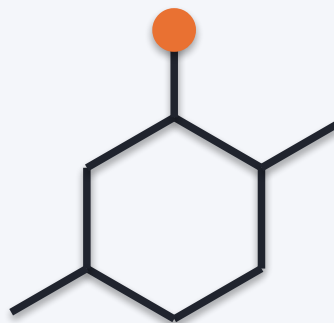
Praski, Adamczyk & Czech, arXiv:2508.06199 (2025) · Jiang et al., J. Cheminform. 2021, 13, 12 · Deng et al., Nat. Commun. 2023, 14, 6395

Activity cliffs: where smoothness is a lie

two molecules, one methyl group apart



$IC_{50} = 8 \text{ nM}$



$IC_{50} = 4,200 \text{ nM}$

Tanimoto similarity 0.95

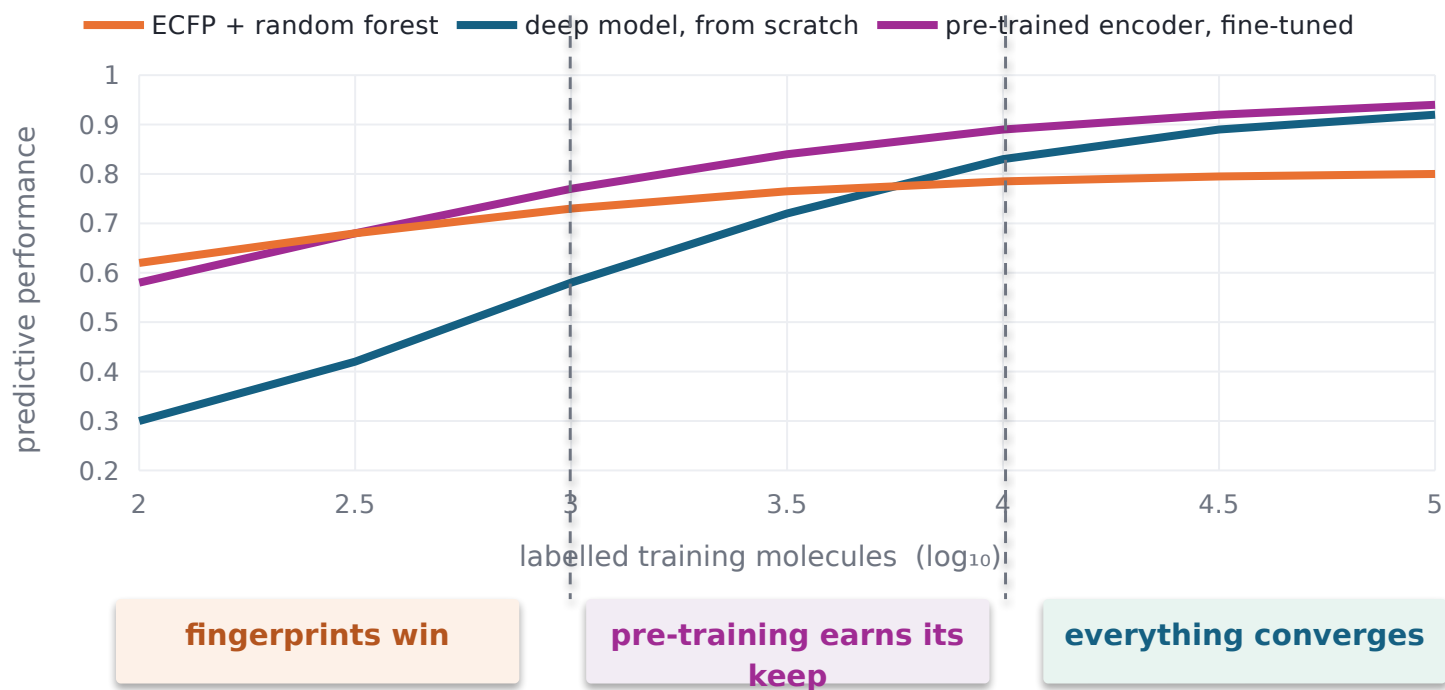
500× potency difference

what breaks

- Smooth embeddings put A and B in the same place.
- Errors on cliff compounds reach 1.4 log units — a 25× error in potency.
- Across 30 targets, SVM on ECFP was the best method overall.
- Transformers pre-trained on 10 M SMILES did not beat LSTMs pre-trained on 36 k.

Smoothness is an assumption, and cliffs violate it. **Report cliff performance separately** — overall RMSE hides exactly the compounds that matter.

So when is a learned representation worth it?



schematic

The shapes are real;
the exact crossover is not.

Published estimates put it
anywhere between 10^3 and 10^4
labelled molecules, and it moves
with label noise, split type and
how narrow your chemistry is.

Measure it on your own data —
that is what the tutorial does.

The shapes are robust; **the crossover is yours to measure**, and it moves with label noise, split type and how narrow your chemistry is.

What to do on Monday

How many labelled examples do you have?

< 100

Do not train a deep model.
ECFP + RF, or a Gaussian process.
Consider frozen pre-trained embeddings
as extra features.

100 - 5,000

This is where pre-training pays.
Frozen embeddings first, then partial
fine-tuning. Always report the ECFP
baseline.

> 10,000

Train from scratch on your own data.
Pre-training becomes a convenience,
not a necessity.

In all three columns: run the baseline first, and split by scaffold, not at random.

Take-home messages

- **Labels are the bottleneck, not molecules.** Everything today is a way of spending unlabelled data to make up for scarce labels.
- **Unsupervised methods find real structure** — reaction classes, the periodic table — but a t-SNE plot tells you about your fingerprint, not about chemistry.
- **The VAE's decoder disappointed; its encoder changed the field.** Pretext task → keep the encoder → reuse it. That is the whole pattern.
- **Validity was the easy metric.** 100% valid is achieved by construction and says nothing about quality, novelty or synthesizability.
- **Pre-training is a prior.** Worth $\sim 3\times$ at a hundred labels, worth nothing at ten thousand, occasionally worth less than nothing.
- **Always run ECFP + random forest, and always split by scaffold.** In the largest independent replication, almost nothing beat that baseline.

Further reading

- **Chemical VAE** — Gómez-Bombarelli et al., ACS Cent. Sci. **2018**, 4, 268
- **SELFIES** — Krenn et al., Mach. Learn.: Sci. Technol. **2020**, 1, 045024 · and the critique: Skinnider, Nat. Mach. Intell. **2024**, 6, 437
- **Pre-training GNNs** — Hu et al., ICLR **2020**, arXiv:1905.12265
- **MolFormer** — Ross et al., Nat. Mach. Intell. **2022**, 4, 1256 · **MolCLR** — Wang et al., Nat. Mach. Intell. **2022**, 4, 279
- **Reaction space** — Schwaller et al., Nat. Mach. Intell. **2021**, 3, 144
- **Transfer for NMR** — Han & Choi, J. Phys. Chem. Lett. **2021**, 12, 3662
- **Do they actually help?** — Deng et al., Nat. Commun. **2023**, 14, 6395 · Praski et al., arXiv:2508.06199 (**2025**) · van Tilborg et al., JCI **2022**, 62, 5938
- **Evaluation guidelines** — Bender et al., Nat. Rev. Chem. **2022**, 6, 428

Questions

Learned Representations

Tutorial · 15:15 · RSC 275

Chemical space maps · frozen vs learned embeddings · the learning-curve crossover

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